

ANKAN PURI

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RESEARCH INTERESTS

Deep Learning Optimization, Edge Computing & Resource-Constrained AI, Computer Vision, Privacy-Preserving Training Protocols, and Digital Twin Edge Networks for Next-Generation (6G) Communications.

EDUCATION

Indian Institute of Information Technology Ranchi

B.Tech in Computer Science and Engineering

Ranchi, India

Aug 2023 – May 2027

- An Institute of National Importance, Govt. of India
- Cumulative Grade Point Average (SGPA): 7.98/10.0

Contai Model Institution (Govt. of WB)

Class XII, WBCHSE (Science Stream)

West Bengal, India

Graduated: May 2023

- Percentage: 78%

RESEARCH EXPERIENCE

Undergraduate Researcher - Medical Image Analysis

Jan 2025 – Present

Computer Vision Lab, Indian Institute of Information Technology Ranchi | Advisor: Dr. Dhiran Kumar Mahto Ranchi, India

- Spearheaded the design and deployment of a highly robust, end-to-end deep learning framework (classification, segmentation, detection, localization) for automated gastrointestinal anomaly screening from WCE video streams.
- Engineered a novel ResNet50-UVNet hybrid architecture, achieving 96.25% diagnostic accuracy in identifying focal structural abnormalities (polyps, bleeding, ulcers) under complex mucosal variability.
- Architected a pixel-precise semantic segmentation pipeline utilizing U-Net; resolved critical image-mask augmentation desynchronization by integrating Albumentations with synchronized transform targets.
- Deployed a low-latency YOLOv8n detection module coupled with a lightweight MobileNet V2 architecture, optimizing computational efficiency for edge-device clinical deployment.
- Diagnosed and mitigated recurrent GPU memory bottlenecks during multi-day Kaggle training iterations by implementing explicit garbage collection, CUDA cache clearing, and JSON-based checkpointing.
- Systematically benchmarked ResNet-50, VGG-16, and EfficientNet-B0 across 10,000+ labeled WCE frames, successfully reducing the false-positive rate by 15% via precise threshold tuning and advanced augmentations.

PUBLICATIONS & MANUSCRIPTS

- A. Puri, D. K. Mahto, et al. "Multi-Pipeline Deep Learning Framework for WCE Lesion Analysis." Manuscript in preparation for the 11th International Conference on Computer Vision & Image Processing (CVIP 2026, Springer LNCS).
- D. K. Mahto, A. Puri, et al. "DO-ResNet50-Based Zero-Watermarking with Hyperbolic Tangent Chaotic Encryption for Securing Medical Data." Under Submission. (Achieved PSNR 39.86 dB, SSIM 0.9867, NC 0.9976).
- A. Puri. "Artificial Intelligence Applications in Ewing Sarcoma: A Review." International Research Journal of Engineering and Technology (IRJET). [Published]

TECHNICAL PROJECTS

Customer Churn Prediction Dashboard | *Python, scikit-learn, Streamlit, Plotly*

Aug 2026

- Developed an end-to-end ML system with a live Streamlit dashboard enabling non-technical teams to instantly evaluate customer risk profiles and estimated revenue at risk.
- Trained a RandomForest classifier on 7,000+ customer records achieving 77% accuracy and 0.84 AUC.
- [Source Code](#) | [Live Demo](#)

Scalable Biometric Attendance Verification | *Python, OpenCV, CNN, KNN*

Nov 2025

- Built an automated facial recognition pipeline extracting 128-dimensional embeddings, attaining 97% accuracy across 30+ subjects under highly variable lighting conditions.

Eye-Controlled Mouse & Clap Detection System | *Python, OpenCV, MediaPipe, PyAudio*

Nov 2025

- Engineered a hands-free cursor control interface utilizing facial mesh estimation; integrated precise blink-event detection mapped dynamically to mouse-click simulations.
- Incorporated real-time audio signal processing using PyAudio and NumPy to detect claps for triggering automated actions (e.g., application launches).
- [Project Preview & Demo](#)

LICENSES & CERTIFICATIONS

- **PCRA Youth Programme** – Petroleum Conservation Research Association, Govt. of India
- **Yuva AI for All** – NASSCOM / IT-ITeS Sector Skills Council (Sep 2026)
- **Cyber Job Simulation** – Deloitte Australia via Forage (Jul 2025)
- **Career Guidance Program** – E-Cell IIT Hyderabad & SkillDunia (Jul 2025)
- **Paper Publication Certificate: AI in Pediatric Bone Tumor Diagnosis** – IRJET (Jun 2026)

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, HTML, CSS

Machine Learning & CV: CNN, ViT, ResNet, U-Net, YOLOv8, Domain Adaptation, Anomaly Detection, RandomForest

Frameworks & Libraries: PyTorch, TensorFlow, Keras, OpenCV, MediaPipe, Albumentations, Scikit-learn, Streamlit, Plotly

Tools & Platforms: Git, GitHub, Google Colab, Kaggle, Jupyter, VS Code, LaTeX